

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

Practical – IX

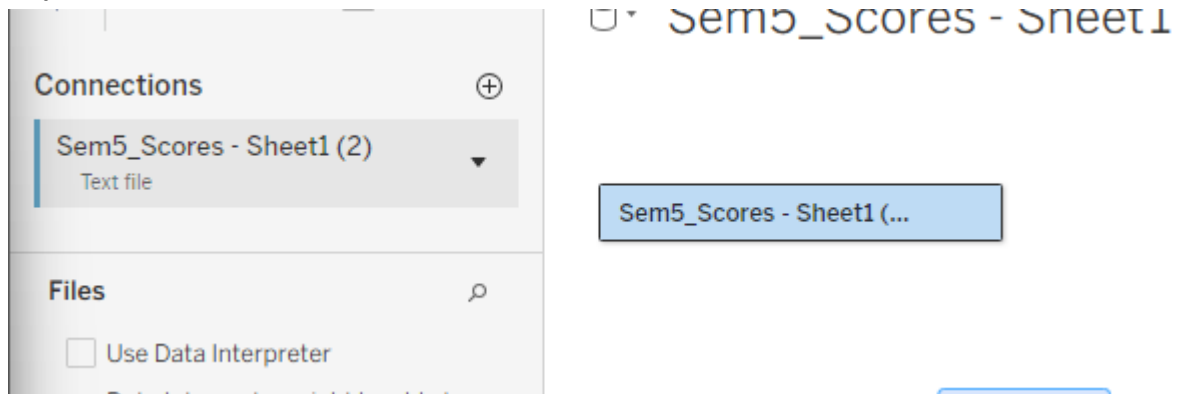
Roll No.: T038	Name: Akansha Pritam Ranade
Class: TYIT	Batch: 1
Date of Assignment: 27-08-2026	Date/Time of Submission: 01-09-2026

Part B – Program

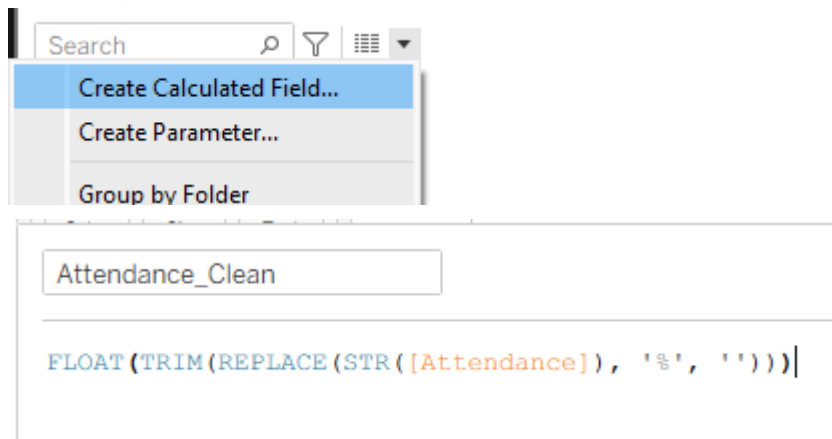
Cleaning and Structuring Data for Analysis

a) Identify and handle missing or inconsistent data

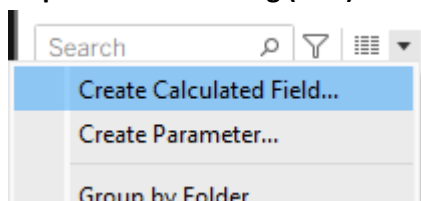
Step 1: Load the Dataset



Step 2: Clean Inconsistent String Fields (Convert Text Percentages to Numeric Measures)



Step 3: Handle Missing (Null) Numeric Values via Imputation



Math_Clean

IFNULL([Math], 0)

Physics_Clean

IFNULL([Physics], 0)

Tables	
Abc	Attendance
Abc	Department
Abc	Full Name
Abc	Student ID
Abc	Measure Names
=#	Attendance_Clean
#	Chemistry
#	Math
=#	Math_Clean
#	Physics
=#	Physics_Clean
#	Programming
#	Sem5_Scores - Sheet1 (2...

b) Convert wide data into tall format
Step 1: Open the Data Source Grid

New Union

New Table Extension

Name Sem5_Scores - Sheet1 (2).csv

Description

No description available.

Fields

Type	Field Name	Physical Table
Abc	Student ID ▼	Sem5_Scores - Sl
Abc	Full Name ▼	Sem5_Scores - Sl
Abc	Department ▼	Sem5_Scores - Sl
#	Math ▼	Sem5_Scores - Sl
#	Physics ▼	Sem5_Scores - Sl
#	Chemistry ▼	Sem5_Scores - Sl
#	Programming ▼	Sem5_Scores - Sl

Data Source

Sheet 1

Step 2: Multi-Select and Pivot Columns

Sem5_Scores - Sheet1 (2)...

11 fields 10 rows

Name Sem5_Scores - Sheet1 (2).csv

Description

No description available.

Fields

Type	Field Name	Physical Table	Rem...
#	Math ▼	Sem5_Scores - Sheet1 ...	Math
#	Physics ▼	Sem5_Scores - Sheet1 ...	Physics
#	Chemistry ▼	Sem5_Scores - Sheet1 ...	Chemi...
#	Programming ▼	Sem5_Scores - Sheet1 ...	Progr...
Abc	Attendance ▼	Sem5_Scores - Sheet1	Atten

Step 3: Rename the Pivoted Fields Tableau replaces the four columns with two consolidated columns at the end of the table:
name the pivot field names and values as Subject and Score respectively

Abc Pivot Subject	# Pivot Pivot Field Values
Chemistry	80
Math	99
Physics	90
Programming	95
Chemistry	80
Math	65

Abc Pivot Subject	# Pivot Score
Chemistry	80
Math	99
Physics	90
Programming	95
Chemistry	80
Math	65
Physics	75

Name: Sem5_Scores - Sheet1 (2/1/2021)

Description

No description available.

Fields

Type	Field Name	Physical Table	Rem...
Abc	Full Name ▼	Sem5_Scores - Sheet1 ...	Full_N...
Abc	Department ▼	Sem5_Scores - Sheet1 ...	Depar...
Abc	Attendance ▼	Sem5_Scores - Sheet1 ...	Atten...
Abc	Subject ▼	Pivot	Pivot ...
#	Score ▼	Pivot	Pivot ...
=#	Attendance_C ▼	Calculation	Calcul...
=#	Math_Clear ! ▼	Calculation	Calcul...
=#	Physics_Cle ! ▼	Calculation	Calcul...

Step 4: Clean and Format the Tall Score Field Because raw Physics contained % symbols, the new combined Score column defaults to text (Abc). Convert it to a numeric measure: Click on score header # -> select number (decimal)

Description

No description available.

Fields

Type

Number (decimal)

✓ Number (whole)

Date & Time

Date

String

Boolean

✓ Default

Geographic Role

Sem5_Scores - Sheet1 (2)....

9 fields 40 rows

40 → rows

Name Sem5_Scores - Sheet1 (2).csv

Description

No description available.

Fields

Type	Field Name	Physical Table	Rem...
Abc	Full Name ▼	Sem5_Scores - Sheet1 ...	Full_N...
Abc	Department ▼	Sem5_Scores - Sheet1 ...	Depar...
Abc	Attendance ▼	Sem5_Scores - Sheet1 ...	Atten...
Abc	Subject ▼	Pivot	Pivot ...
#	Score ▼	Pivot	Pivot ...
-#	Attendance_C ▼	Calculation	Calcul...
-#	Math_Clear ! ▼	Calculation	Calcul...
-#	Physics_Ck ! ▼	Calculation	Calcul...

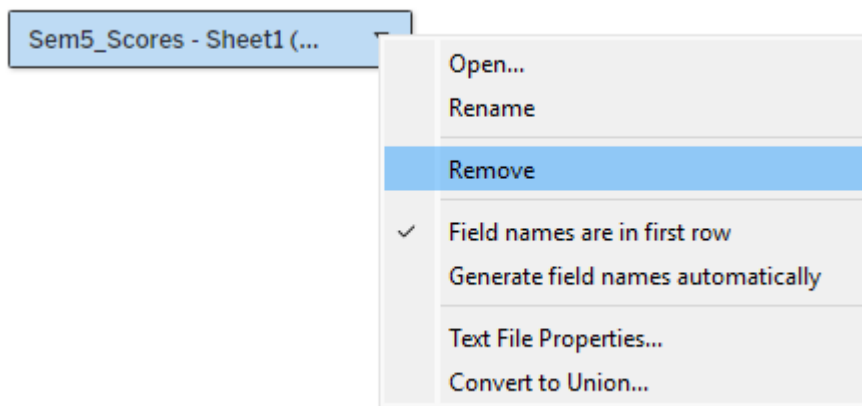
	# Pivot Score	# Calcula Atten
try	80.0000	
	99.0000	
	90.0000	
nming	95.0000	
try	80.0000	
	65.0000	
	75.0000	
nming	85.0000	
try	86.0000	
	null	
	82.0000	

Clean_Score

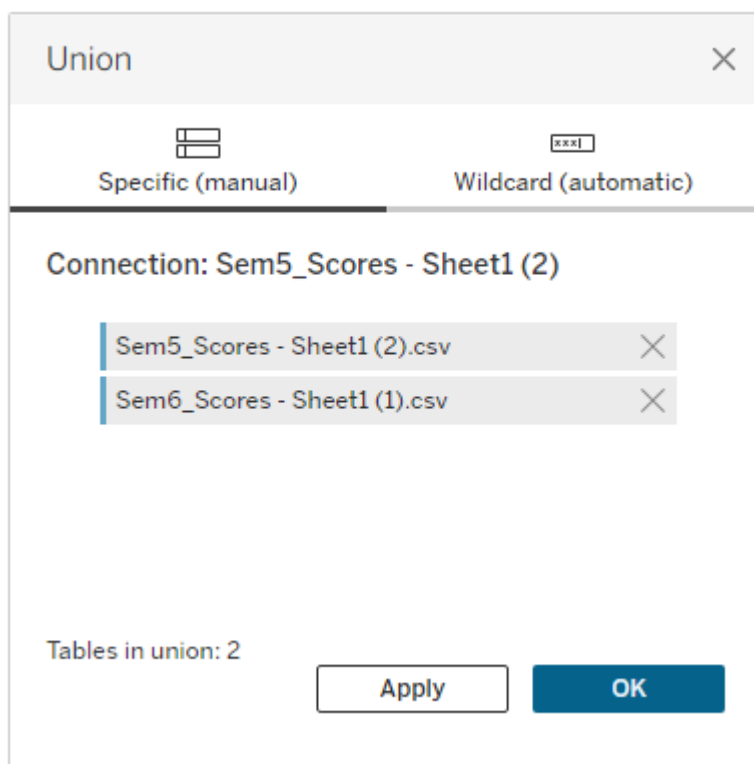
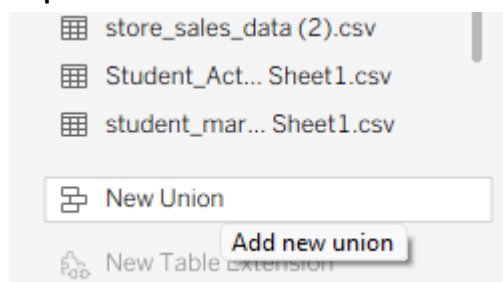
`IFNULL(FLOAT(REPLACE(STR([Score]), '%', '')), 0)`

c) Perform union of multiple files

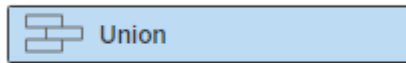
Step 1: Open the Data Source Canvas



Step 2: Create a Multi-File Union



Extract will o



Logical Table: Union
Unioned Tables: Sem5_Scores - Sheet1 (2).cs...
(1 More)

Step 3: Verify the Merged Data Grid



Union	13 fields 20 rows	20	→ rows
Name Union	<	Abc	▼
Description 		Union	
		Student_ID	

Union

13 fields 20 rows

20 → rows

Name Union

Description No description available.

Fields

Type	Field Name	Physical Table	Remote F...
Abc	Student_ID	Union	Student_ID
Abc	Full_Name	Union	Full_Name
Abc	Department	Union	Department
...	...	...	...

Abc

Union

Table Name

Sem5_Scores - Sheet1 (

Sem5_Scores - Sheet1 (

Sem5_Scores - Sheet1 (

Sem5_Scores - Sheet1 (

Sem5_Scores - Sheet1 (

Sem5_Scores - Sheet1 (

Sem5_Scores - Sheet1 (

Step 4: Create a Clean Semester Identifier Dimension

Semester

IF CONTAINS([Table Name], "Sem1") THEN "Semester 1"
 ELSEIF CONTAINS([Table Name], "Sem2") THEN "Semester 2"
 ELSE "Other"
 END

The calculation is valid.

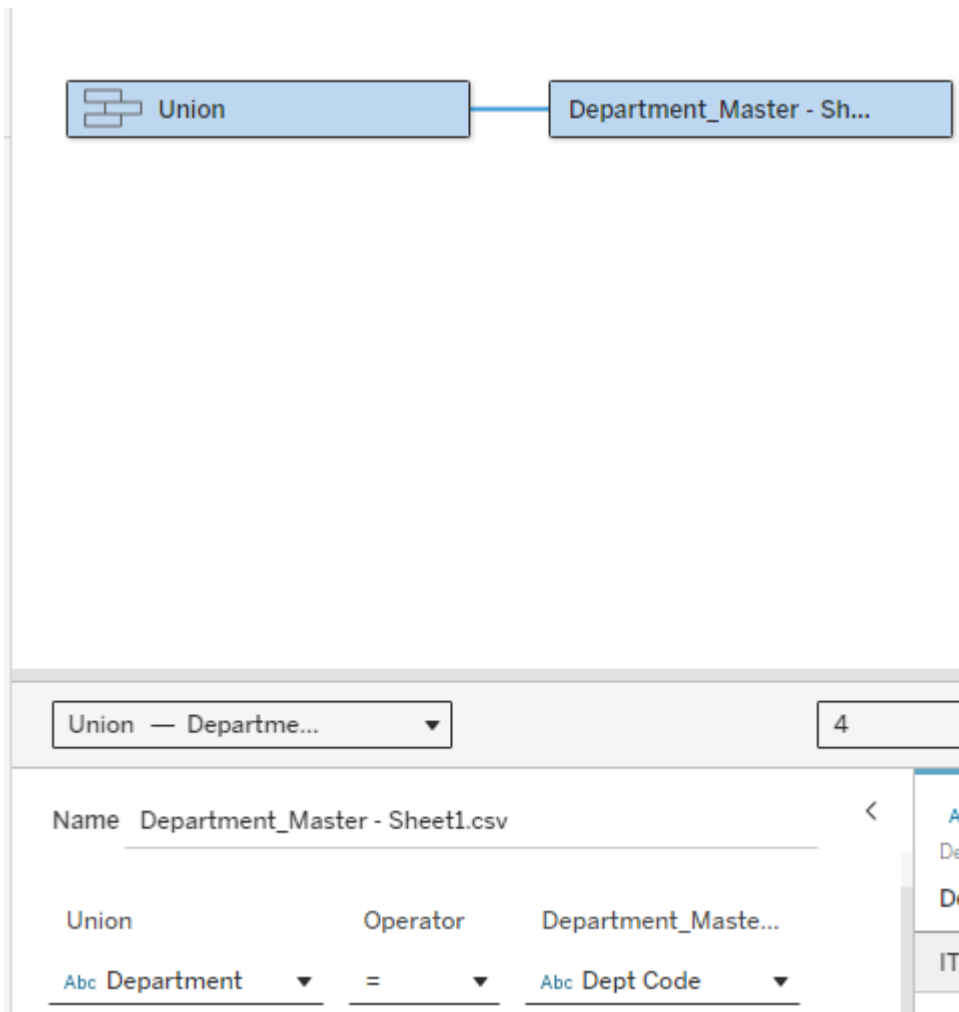
Apply

OK

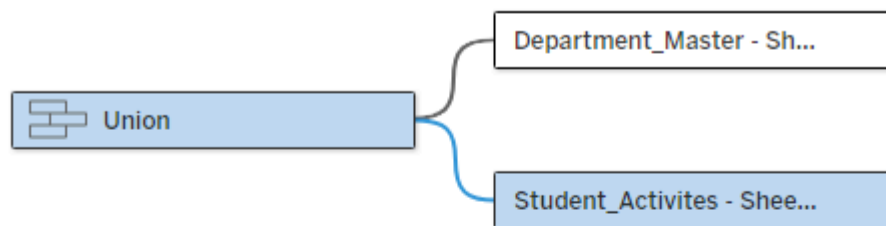
d) Execute cross-database joins

Step 2: Add and Link Department_Master.csv

Go to data source-> drag department_master->if relationship is not build then on left side click on department and on right side click o dept_code



Step 3: Add and Link Student_Activities.csv



Union — Student_A... 10 →

Name Student_Activites - Sheet1.csv

How do relationships differ from joins? [Learn more](#)

Union	Operator	Student_Activites - ...
Abc Student_ID	=	Abc Student ID (Stui

Step 4: Verify in Sheet 1

Union	
Abc	Attendance
Abc	Department
Abc	Full_Name
=Abc	Semester
Abc	Student_ID
Abc	Table Name
=#	Attendance_Clean
#	Chemistry
#	Math
=#	Math_Clean
#	Physics
=#	Physics_Clean
#	Programming
#	Union (Count)

e) Restructure data for better visualization

Step 1: Calculate Total Marks and Scaled Percentage

Total_Score

$$[\text{Math_Clean}] + [\text{Physics_Clean}] + [\text{Physics_Clean}] + [\text{Programming}]$$

Overall_Percentage

$$[\text{Total_Score}] / 400.0$$

Abc	Department
Abc	Full_Name
=Abc	Semester
Abc	Student_ID
Abc	Table Name
=#	Attendance_Clean
#	Chemistry
#	Math
=#	Math_Clean
=#	Overall_Percentage

Convert to Dimension
Change Data Type
Default Properties
Geographic Role
Group by
Folders
Replace References...
Describe...

Comment...
Color...
Number Format...
Aggregation
Total using

Default Number Format [Overall_Percentage]

Automatic
Number (Standard)
Number (Custom)
Currency (Standard)
Currency (Custom)
Scientific
Percentage
Custom

Percentage
Decimal places:
1

Step 2: Create Categorical Performance Bands (Grading Tiers)

Performance_Band

```

IF [Overall_Percentage] >= 0.85 THEN "Distinction"
ELSEIF [Overall_Percentage] >= 0.75 THEN "First Class"
ELSEIF [Overall_Percentage] >= 0.60 THEN "Second Class"
ELSE "Needs Academic Support"
END

```

Step 3: Create an Attendance Health Flag

Attendance_Status ✕

IF [Attendance_Clean] >= 75 THEN "Eligible (Good Standing)"
ELSE "Shortage Warning (<75%)"
END

Step 4: Build the Student Performance Scorecard View

Pages

Filters

Marks

Automatic

Color

Size

Text

Detail

Tooltip

Performance_...

ATTR(Attenda..

SUM(Overall_..

Columns

Student ID (Stude..

Full_Name

Rows

Semester

Sheet 1

	Student ID (Student Activi..		
	T101	T102	T103
Semester	Akansha Ranade	Priyal Verma	Rohan Gupta
Other	190.0%	...	...

Performance_Band

Distinction

First Class

Needs Academic Sup..

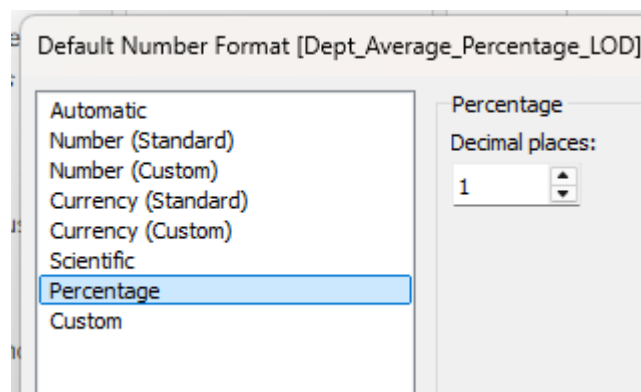
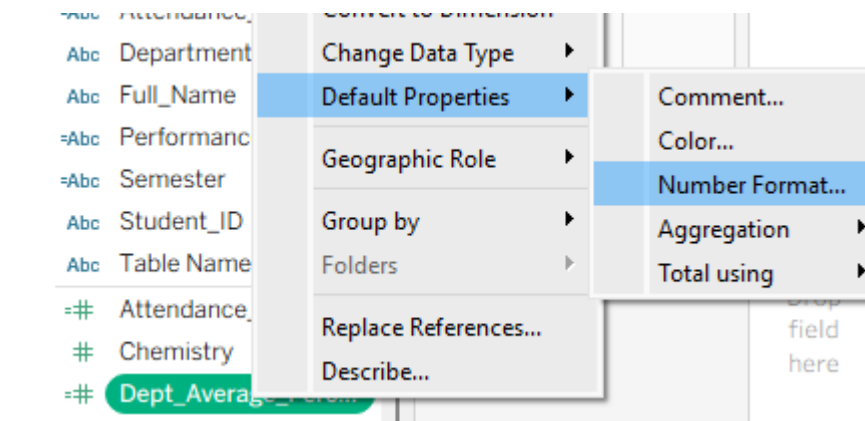
Second Class

f) Handle different levels of detail in datasets

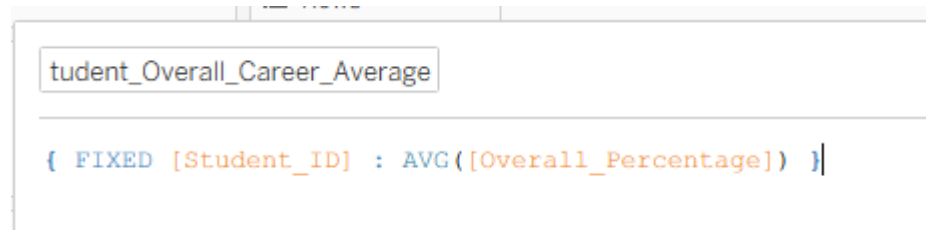
Step 1: Compute Department-Level Benchmark using {FIXED} LOD

Dept_Average_Percentage_LOD

```
{ FIXED [Department] : AVG([Overall_Percentage]) }
```



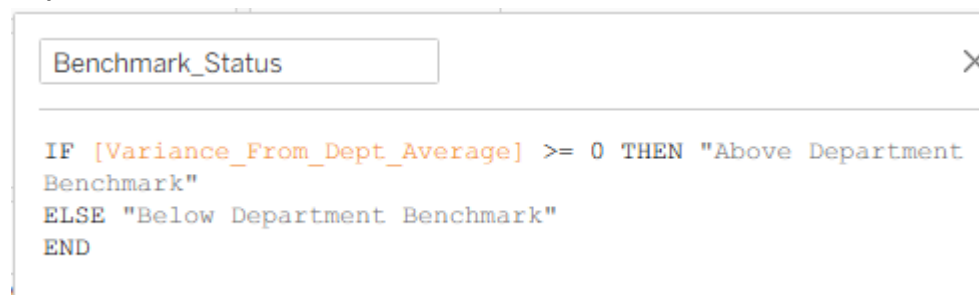
Step 2: Compute Student Multi-Semester Average using {FIXED} LOD
Student_Overall_Career_Average



Step 3: Calculate Performance Variance (Student vs. Department Benchmark)



Step 4: Create a Benchmark Indicator Dimension



Step 5: Build the LOD Benchmark Visualization

Pages

Filters

Marks

Automatic

Color

Size

Text

Detail

Tooltip

Benchmark_St..

SUM(Dept_Av..

SUM(Overall_..

Columns

Semester

Rows

Department

Full_Name

Sheet 2

Department	Full_Name	Semester	Other
CS	Rahul Nair	91.8%	71.3%
	Ananya Singh	90.5%	69.3%
ECE	Priyal Verma	85.3%	75.0%
	Aditya Rao	87.5%	26.3%
IT	Akansha Ranade	190.0%	
	Kavya Reddy	171.0%	
	Rohan Gupta	91.0%	63.0%
	Pooja Iyer	85.3%	77.3%
MECH	Sneha Patel	91.0%	74.5%
	Vikram Joshi	91.3%	64.8%

Benchmark_Status

Above Department ..

Below Department B.

Benchmark_Status

Above Department ..

Below Department B..

g) Apply advanced data cleaning techniques

Step 1: Clean Leading/Trailing Whitespace and Inconsistent Name Casing

Full_Name_Clean

×

```

UPPER(LEFT(TRIM([Full_Name]), 1)) +
LOWER(MID(TRIM([Full_Name]), 2, FIND(TRIM([Full_Name]), " ") -
1)) +
" " +
UPPER(MID(TRIM([Full_Name]), FIND(TRIM([Full_Name]), " ") +
1)) +
LOWER(MID(TRIM([Full_Name]), FIND(TRIM([Full_Name]), " ") +
LEN(TRIM([Full_Name]))))

```

The calculation is valid.

Apply

OK

Full_Name_Clean

×

```

TRIM(LEFT(TRIM([Full_Name]), 1)) +
TRIM(MID(TRIM([Full_Name]), 2, FIND(TRIM([Full_Name]), " ") -
1)) +
" " +
TRIM(MID(TRIM([Full_Name]), FIND(TRIM([Full_Name]), " ") + 1,
FIND(TRIM([Full_Name]), " ") + 2,
LEN(TRIM([Full_Name]))))

```

Step 2: Parse First Name and Last Name using SPLIT()

First_Name

TRIM(SPLIT([Full_Name], " ", 1))

Last_Name

TRIM(SPLIT([Full_Name], " ", -1))

Step 3: Split Composite Department Codes (Department)

Dept_Branch

```
SPLIT([Department], "-", 1)
```

Selection_ID

```
SPLIT([Department], "-", 2)
```

Step 4: Create a Clean Composite Student Label

Student_Display_Label ✕

```
[Student_ID] + " - " + [Full_Name_Clean] + " (" + [Dept_Bran
```

Student_Display_Label

```
" - " + [Full_Name_Clean] + " (" + [Dept_Branch] + ") "
```

Step 5: Verify in the Final Dashboard View

Pages

ColumnsSemester

RowsStudent_Display_Lab..

Filters

Dept_Branch

Selection_ID

Marks

Automatic

Color

Size

Text

Detail

Tooltip

Performance_..

SUM(Overall_..

Sheet 3

Student_Displ..	Semester
	Other
T101 - Akansha Ranade (IT)	190.0%
T102 - Priyal Verma (ECE)	85.3% 75.0%
T103 - Rohan Gupta (IT)	91.0% 63.0%
T104 - Sneha Patel (MECH)	91.0% 74.5%
T105 - Aditya Rao (IT)	87.5% 26.3%

Performance_Band

Distinction

First Class

Needs Academic Sup..

Second Class